

Phase Information Theory: Information, Limits, and Phase Distinguishability

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Abstract

We develop **Phase Information Theory (PIT)**, a foundational reformulation of information theory in which information is not an abstract symbol count, a property of physical carriers, nor an independent primitive of nature, but rather a precise measure of **distinguishability between admissible phase configurations**. In the framework established across Papers 1-11 of the Phase Theory series, the phase field Φ defined on the admissible configuration space Ω_{adm} constitutes the sole ontological substrate. The present paper shows that information arises as a derived structural property of this substrate: specifically, from the separability of configurations under global admissibility constraints, without appealing to wavefunctions,

probability axioms, Hilbert spaces, or symbolic encoding conventions.

We formally define the phase distinguishability functional $D(\Phi_1, \Phi_2)$ and derive from it a log-cardinality measure $I_{\text{info}} = \log|D|$ that subsumes Shannon entropy, von Neumann entropy, and algorithmic complexity as limiting cases parameterized by phase overlap depth δ . Classical bits emerge at $\delta = 0$ (fully separated basins); quantum bits emerge at $\delta = 1$ (fully overlapping basin neighborhoods); hybrid and analog information structures occupy the continuous interior. Fundamental information bounds are then derived entirely within the phase-structural language: the Phase No-Cloning Theorem (Theorem 12.1) follows from the non-separability of global phase constraints; a modified Landauer bound is obtained from the energetic cost of phase basin collapse; and phase information capacity C_{phase} is bounded by coherence length, admissibility curvature, and topological protection density.

We further argue that canonical information paradoxes—including black hole information loss, Landauer irreversibility, and the no-cloning prohibition—dissolve into structural statements about phase configuration geometry. A preview of the black hole treatment (Paper 13) is included. Taken together, these results establish the central thesis of Paper 12: *information is not stored in reality; reality is structured so that distinctions can exist*. This paper is self-contained but presupposes familiarity with the phase substrate formalism [PT1], the admissibility calculus [PT2, PT3], the elimination of intrinsic randomness [PT5], the constraint-restructuring account of measurement [PT6], and the phase-structural derivations of matter and fields [PT7-PT9].

1. The Ontological Status of Information

Few concepts have proven simultaneously more indispensable and more philosophically elusive in modern physics than **information**. It appears in the Second Law of Thermodynamics, in the foundations of quantum mechanics, in

the statistical mechanics of black holes, and in the computational complexity of physical processes. Yet when one demands of any given framework a precise answer to the question—*what, ontologically, is information?*—the frameworks typically demur. Information, it seems, is invoked as freely as energy and as loosely as entropy, without the discipline of a foundational definition that does not already presuppose the very concepts it is meant to illuminate.

Three dominant positions compete in the contemporary literature. Shannon’s original formulation treats information as an abstract, substrate-independent quantity: a statistical measure of the reduction of uncertainty in a receiver upon receiving a message from a sender operating over a noisy channel [1]. On this view, information is defined purely by the probability distribution over an alphabet; the physical implementation is irrelevant. This position is mathematically powerful and operationally clean, but it renders information “about” physical states without being “in” the physical world in any explanatory sense. It cannot, therefore, serve as a foundation for physical reasoning about information bounds, information loss, or the informational content of quantum states.

The quantum information view, developed systematically by Nielsen and Chuang [2] and underwritten by the operator-algebraic formalism of Hilbert space quantum mechanics, holds that information is inherently *physical* and *state-based*: it is carried by quantum states, bounded by the Holevo capacity [3], and subject to operational constraints including no-cloning [4, 5] and no-signaling. This view has been extraordinarily productive, but it purchases its precision at the price of taking the quantum state—and thus the wavefunction—as a primitive. Under Phase Theory, in which the wavefunction is eliminated in favor of phase-structural constraint geometry [PT6], the quantum information framework cannot serve as a foundation; it must instead emerge from the deeper structure.

The computational view, developed in the tradition of Kolmogorov [6], Chaitin, and algorithmic information theory, treats information as a resource: the length of the shortest program sufficient to reproduce a string. This approach successfully connects information to computational complexity and provides a universal notion of description length that is substrate-independent in principle. But it imports the notion of a universal computing machine as a primitive, which is itself a physical assumption requiring grounding. Moreover, Kolmogorov complexity is uncomputable in general, making it unsuitable as a foundational physical quantity.

Wheeler’s celebrated “**it from bit**” [7] represents the most radical attempt to make information foundational: he proposed that physical reality is constituted by answers to binary observational questions, so that “every it—every particle, every field of force, even the spacetime continuum itself—derives its function, its meaning, its very existence... from the apparatus-elicited answers to yes-or-no questions.” But this proposal inverts the correct explanatory order. It makes the *act of observation* ontologically primary, when observation is itself a physical process that requires explanation. Under Phase Theory [PT6], observation (measurement) is a constraint restructuring event in phase space—a derived, not a primitive, notion. Wheeler’s it-from-bit cannot accommodate this inversion.

Zurek’s existential interpretation and his program of quantum Darwinism [8, 9] place information at the center of the emergence of classical reality: pointer states are selected by environmental robustness, and the redundant imprinting of pointer-state information into the environment constitutes objective classical facts. This is conceptually rich, but it grounds information in decoherence and environmental selection, which are themselves features of quantum mechanics assumed as a starting point. Phase Theory eliminates the environment-as-separate-entity picture [PT8] and must therefore reconstruct any Darwinist story from phase-structural principles alone.

Rovelli’s relational quantum mechanics [10] proposes that quantum states are relational rather than absolute: the state of a system is always relative to another system, and information is precisely what one system “has” about another. This is structurally resonant with PIT’s emphasis on relational distinguishability. However, Rovelli’s framework still employs quantum states and Hilbert-space formalism as its mathematical substrate, and it does not provide a derivation of why relational information obeys the constraints it does (no-cloning, Holevo bound, etc.). PIT provides exactly this derivation from phase geometry.

The situation from the standpoint of Phase Theory is now precise: Papers 1–11 of this series have progressively eliminated the need for intrinsic randomness [PT5], wavefunction collapse [PT6], fundamental particles [PT7], fields as primitives [PT8, PT9], and thermodynamic entropy as an independent axiom [PT8].¹ If all of these are derived features of the phase substrate, then information—which connects all of them—cannot be a primitive either. It must be derivable from phase structure alone. The purpose of this paper is to carry out that derivation with full rigor and to show that the resulting theory reproduces and extends all the major results of classical and quantum information theory, while resolving the standing paradoxes that those frameworks generate.

¹ The series currently comprises: [PT1] Phase Ontology and the Structure of the Admissible Configuration Space; [PT2] The Admissibility Calculus I: Constraint Operators and Global Consistency; [PT3] The Admissibility Calculus II: Boundary Geometry and Phase Curvature; [PT4] Dynamics Without Trajectories: Phase Flow and Temporal Ordering; [PT5] The Elimination of Intrinsic Randomness; [PT6] Measurement as Constraint Restructuring; [PT7] Matter as Phase Condensation; [PT8] Thermodynamics, Entropy, and Phase Basin Statistics; [PT9] Fields as Admissibility Gradients; [PT10] The Phase Derivation of Quantum

Mechanics; [PT11] Spacetime Geometry as Phase Constraint Topology. The present paper, [PT12], introduces Phase Information Theory.

2. Phase Contains No Symbols

Before a positive theory of phase information can be constructed, a negative result must be firmly established: the phase field itself carries no intrinsic symbolic content. This may appear obvious, but its precise formulation has nontrivial consequences for the entire framework.

In Phase Theory, the fundamental ontological entity is the **phase field** Φ , defined as a scalar (or, in higher-rank generalizations, a vector or tensor) field on the phase substrate Σ [PT1]. The field Φ takes values in a continuous space and is subject to global admissibility constraints $G_k[\Phi] = 0$. Nothing in this formalism assigns a label, a letter, a bit, or a symbol to any particular value of Φ . There is no intrinsic correspondence between $\Phi(x) = 0.7$ and the letter “a,” between $\Phi(x) = -1.2$ and the bit “0,” or between any phase configuration and any symbolic encoding whatsoever. Phase values are not drawn from an alphabet. Phase does not speak.

Contrast this with conventional information carriers. A spin-1/2 particle carries information by a *convention*: we agree that spin-up represents “1” and spin-down represents “0.” A voltage rail carries information by a convention: voltages above threshold are “high” (logical 1), voltages below threshold are “low” (logical 0). A photon polarization carries information by a convention: horizontal polarization represents one bit-value, vertical the other. In every case, the physical property is continuous and convention-independent; the informational content is discrete and convention-dependent. Information, on these frameworks, is a projection of physical states onto a symbolic space by an external interpretive act.

Phase Theory eliminates the need for such external interpretive acts by deriving information from the *relational structure* of phase configurations—from their mutual distinguishability under admissible transformations—rather than from any local property of $\Phi(x)$. This elimination is captured formally as follows.

Theorem 2.1 (Phase Symbol Neutrality)

Let Φ be the phase field defined on the admissible configuration space Ω_{adm} as specified in [PT1]. Let I be any information quantity defined as a function of local phase values alone: $I = f(\Phi(x_1), \Phi(x_2), \dots, \Phi(x_n))$ for any finite collection of points $\{x_i\} \subset \Sigma$. Then I is invariant under all global admissible transformations $T \in T_{adm}$ that preserve global constraints while permuting local values. Consequently, I cannot distinguish between configurations that differ only in their global relational structure. No genuine information quantity I can be defined from local phase values alone.

Proof.

Let $T \in T_{adm}$ be a global admissible transformation (as defined in [PT2]). By definition, T preserves the admissibility conditions: $G_k[T(\Phi)] = 0$ for all k whenever $G_k[\Phi] = 0$. Suppose $I = f(\Phi(x_1), \dots, \Phi(x_n))$ for local points $\{x_i\}$. Then $I(T(\Phi)) = f(T(\Phi)(x_1), \dots, T(\Phi)(x_n))$. For a transformation T that is a global reparametrization of Φ by any admissible constant shift or scale—a subgroup of T_{adm} that preserves all G_k —we have $T(\Phi)(x) = \Phi(x) + c$ for all x , where c is a global constant satisfying the linearized constraint equations. Then $I(T(\Phi)) = f(\Phi(x_1) + c, \dots, \Phi(x_n) + c) \neq f(\Phi(x_1), \dots, \Phi(x_n))$ in general. Hence any function of local values is either not invariant under the full group T_{adm} or is constant (uninformative) on Ω_{adm} . In either case it fails to provide a physically meaningful, transformation-covariant information measure. $\square$

The corollary is immediate and important: all information-bearing distinctions in Phase Theory must arise from **relations between phase configurations** rather than from any internal property of Φ at a point. This shifts the foundational question from “what does the local phase value encode?” to “which configurations can be distinguished from which others by admissible processes?” It is precisely this relational, distinguishability-based conception of information that the remainder of this paper develops.

This result also has a philosophical corollary. Phase Theory is not merely claiming that symbols are conventional (a point everyone already accepts); it is claiming that the *structure from which information arises*—the admissibility geometry of Ω_{adm} —is prior to and independent of any symbolic encoding. Information is not a property of phase values; it is a property of phase *separations*, as measured by the distinguishability functional introduced in the next section.

3. Information as Phase Distinguishability

3.1 Admissible Configuration Space

We begin by establishing the formal setting with precision [PT1, PT2]. Let Σ denote the phase substrate, a smooth manifold of dimension n equipped with a connection compatible with global constraint geometry. The **phase field** $\Phi : \Sigma \rightarrow V$ is a map into a value space V (typically $V = \mathbb{R}$ or $V = \mathbb{R}^m$ for some m). The set of all phase fields on Σ is denoted Ω .

The **admissible configuration space** Ω_{adm} is the subset of Ω satisfying all global consistency constraints:

$$\Omega_{\text{adm}} = \{ \Phi \in \Omega : G_k[\Phi] = 0 \quad \forall k \in K \} \quad (12.1)$$

where K is an index set (possibly infinite) and each $G_k : \Omega \rightarrow \mathfrak{R}$ is a **global constraint operator** that may be nonlinear in Φ . The operators G_k encode all physical laws of the phase substrate: field equations, topological boundary conditions, consistency requirements at all scales [PT2, PT3]. Crucially, the constraints are *global*: each $G_k[\Phi]$ depends on Φ across all of Σ , not merely at a point.

The **admissible transformation group** T_{adm} is the group of bijections $T : \Omega_{\text{adm}} \rightarrow \Omega_{\text{adm}}$ that preserve all constraints:

$$T \in T_{\text{adm}} \Leftrightarrow G_k[T(\Phi)] = 0 \quad \forall k, \forall \Phi \in \Omega_{\text{adm}} \quad (12.2)$$

This group captures all physically realizable processes: dynamics, interactions, measurements, and computations are all elements of T_{adm} . Any process not in T_{adm} is physically inadmissible by definition.

3.2 Distinguishability Metric

We now introduce the central object of Phase Information Theory. Two configurations $\Phi_1, \Phi_2 \in \Omega_{\text{adm}}$ are said to be **T_{adm} -equivalent** if there exists a $T \in T_{\text{adm}}$ such that $T(\Phi_1) = \Phi_2$. This defines an equivalence relation $\sim$ on Ω_{adm} , and the quotient $\Omega_{\text{adm}}/\sim$ is the space of physically distinct configurations.

The **phase distinguishability functional** $D : \Omega_{\text{adm}} \times \Omega_{\text{adm}} \rightarrow \{0, 1\}$ is defined as:

$$D(\Phi_1, \Phi_2) = 1 \Leftrightarrow \nexists T \in T_{\text{adm}} : T(\Phi_1) = \Phi_2 \quad (12.3)$$

and $D(\Phi_1, \Phi_2) = 0$ otherwise. Two configurations are **distinguishable** if and only if no admissible transformation maps one to the other; they are **indistinguishable** if and only if they lie in the same T_{adm} -orbit. Note that D is

not a metric in the usual sense (it is not real-valued in the graded case), but it generates a partition of Ω_{adm} into orbits. A **mutually distinguishable set** $M \subset \Omega_{\text{adm}}$ is a subset such that $D(\Phi_i, \Phi_j) = 1$ for all $i \neq j$ in M .

For continuous phase spaces, the binary distinguishability functional is refined to a graded version $D_g : \Omega_{\text{adm}} \times \Omega_{\text{adm}} \rightarrow [0, 1]$, where $D_g(\Phi_1, \Phi_2)$ measures the “minimum admissible transformation cost” of mapping Φ_1 to Φ_2 , normalized by the maximum such cost in Ω_{adm} . The binary case is recovered in the limit of large basin separation.

3.3 Formal Definitions

Definition 12.1 (Phase Distinguishability)

*Let $\Phi_1, \Phi_2 \in \Omega_{\text{adm}}$. The configurations Φ_1 and Φ_2 are **phase-distinguishable** if and only if no element of T_{adm} maps one to the other. The **maximally distinguishable set** $D \subset \Omega_{\text{adm}}$ is a maximal mutually distinguishable subset of Ω_{adm} , i.e., a subset D such that $D(\Phi_i, \Phi_j) = 1$ for all $i \neq j$ in D , and no element $\Psi \in \Omega_{\text{adm}} \setminus D$ can be added to D while preserving this property.*

Definition 12.2 (Phase Information)

*The **phase information** I_{info} of a system with admissible configuration space Ω_{adm} and maximally distinguishable set D is:*

$$I_{\text{info}} = \log_2 |\mathcal{D}|$$

|D|

(12.4)

where $|D|$ is the cardinality of the maximally distinguishable set (or, in the continuous case, the volume of D measured by the appropriate admissibility measure μ_{adm}). Phase information is measured in **phase-bits** (pbits), which reduce to Shannon bits in the classical limit.

We now derive the reduction to Shannon entropy. Suppose that the phase substrate admits a coarse-graining into N distinct basins $B_1, B_2, \dots, B_N$, where each basin is a connected component of Ω_{adm} that is invariant under a subgroup $T_{\text{adm}}^{(i)} \subset T_{\text{adm}}$. Assign probability p_i to basin B_i via the admissibility measure: $p_i = \mu_{\text{adm}}(B_i) / \mu_{\text{adm}}(\Omega_{\text{adm}})$. Then the Shannon entropy $H = -\sum_i p_i \log_2 p_i$ is recovered as the expected phase information under the coarse-grained measure:

$$\begin{aligned} H(\{p_i\}) &= -\sum_{i=1}^N p_i \log_2 p_i = \log_2 N - K_B \\ D_{\text{KL}}(\mathbf{p} \parallel \mathbf{u}) \end{aligned} \quad (12.5)$$

where $\mathbf{u}$ is the uniform distribution over N basins and D_{KL} is the Kullback-Leibler divergence [11]. In the limit of uniform basin measure ($p_i = 1/N$ for all i), $H = \log_2 N = I_{\text{info}}$. Thus Shannon entropy is the special case of phase information in which all distinguishable basins are equiprobable and coarse-graining is exact.

3.4 Information as Availability

A crucial conceptual consequence of Definition 12.2 is that phase information is not a *stored quantity* but a *structural availability*. The phase substrate Ω_{adm} supports a certain number of mutually distinguishable configurations; that number, in logarithmic units, *is* the information. Information does not flow; it

does not accumulate; it does not degrade per se. What changes over time is the geometry of Ω_{adm} and the action of T_{adm} upon it, which collectively determine how many distinct phase states are distinguishable by what classes of processes.

This is a fundamental departure from both the Shannon and quantum information pictures. In Shannon theory, information is a property of a probability distribution, which is a representation of an observer's epistemic state. In quantum information theory, information is a property of a quantum state, which is an objective (if non-classical) physical entity. In PIT, information is a property of the *global geometry of the admissible configuration space*. It is not assigned to a state; it is computed from the structure of state-space itself. This shift from state to space is the key ontological innovation of Phase Information Theory.

4. Information Is Not Physical Substance

A persistent and problematic tendency in the physics of information is to treat information as though it were a conserved substance: something that can flow, be stored, be erased, be transmitted, and be lost. Wheeler's it-from-bit elevates this tendency to a metaphysical principle. Even more sober treatments speak of "information current," "information loss," and "information scrambling" in ways that implicitly assign substance-like properties to information. Phase Theory requires us to scrutinize and ultimately reject these locutions as category errors.

Proposition 4.1 (Information is not a conserved current): There is no vector field J^μ on the phase substrate Σ such that $\partial_\mu J^\mu = 0$ and J^μ represents the flow of phase information I_{info} in all physically admissible processes.

Proof.

Suppose such a J^μ exists. Then $I_{\text{info}} = \int_\Sigma J^0 d\Sigma$ is conserved in the sense that $d I_{\text{info}}/dt = 0$. But $I_{\text{info}} = \log_2 |D|$, where $|D|$ is the cardinality of the maximally distinguishable set. Consider a process $T \in T_{\text{adm}}$ that restructures the admissibility boundary geometry while preserving $|\Omega_{\text{adm}}|$. Such a process may increase or decrease $|D|$ without violating any global constraint, because $|D|$ depends on the T_{adm} -orbit structure, not on $|\Omega_{\text{adm}}|$ alone. Hence $d I_{\text{info}}/dt \neq 0$ in general under admissible processes, contradicting the assumption that J^μ is a conserved current for I_{info} . $\square$

Equally, phase information is not energy (it is dimensionless and can change without energy exchange in the presence of constraint rearrangements at zero net work), not matter (it has no mass, charge, or other Noether charge), and not a Noether charge itself (it is not derived from any continuous symmetry of the phase action via Noether's theorem [PT4]).

This does not mean that information-talk is meaningless in PIT; it means that it must be translated into precise phase-structural language. The table below provides the canonical translation dictionary.

Conventional Language	PIT Reformulation	Phase-Structural Mechanism
"Information flows from A to B"	A temporal sequence of distinguishability relations in which an observer at B gains phase-distinguishability access not previously held	Global phase constraint reconfiguration extends the effective distinguishability range of B-local probes
"Information is stored in a memory"	A phase basin attractor persists over time, maintaining its distinguishability from all other basins	Basin stability under T_{adm} : the attractor remains in its equivalence class across all dynamical trajectories
"Information is erased"	Two previously distinguishable phase basins are merged into one by an admissible process; $ D $ decreases by at least 1	Basin collapse: the admissibility boundary separating two basins is removed by constraint restructuring

Conventional Language	PIT Reformulation	Phase-Structural Mechanism
“Information is lost”	External distinguishability $ D_{\text{external}} $ decreases while total $ \Omega_{\text{adm}} $ is preserved; internal distinguishability is inaccessible to external probes	Phase saturation at a system boundary converts external distinguishability into internal phase structure
“Information is transmitted”	Phase constraint restructuring allows a probe at a remote location to resolve distinguishability that previously required local access	Propagation of admissibility boundary geometry at the phase propagation speed
“Quantum information is entangled”	Two sub-regions of Ω_{adm} have non-factorizable admissibility constraints; their orbit structures are globally correlated	Non-separability of global constraint operators G_k across sub-region factorization

The entries in this table are not mere terminological substitutions; they are derivational obligations. In each case, PIT must show (and does show, across this paper and the series) that the phase-structural mechanism explains all the empirical phenomena that the conventional language was tracking, without importing the ontological commitments (information as substance, information as conserved charge) that generate paradoxes.

5. Classical Bits as Phase Limits

Having established the general framework, we now derive classical information structures as limiting cases of phase distinguishability geometry. The derivation shows that classical bits—far from being foundational—are a highly idealized limit of phase structure that obtains only when basin separation is large relative to all relevant phase fluctuation scales.

Define a **phase basin** $B_\alpha \subset \Omega_{\text{adm}}$ as a connected, T_{adm} -invariant subset of the admissible configuration space that is an attractor under the phase dynamics

of [PT4]. The **basin depth** λ_α is the minimum T_{adm} -transformation cost required to exit B_α :

$$\lambda_\alpha = \min_{T \in T_{\text{adm}}, \Phi \in B_\alpha, T(\Phi) \notin B_\alpha} \quad (12.6)$$

Definition 12.3 (Classical Bit in PIT)

A **phase classical bit** is a pair of phase basins $\{B_0, B_1\} \subset \Omega_{\text{adm}}$ such that:

- (i) $D(B_0, B_1) = 1$ (the basins are distinguishable);
- (ii) $\lambda_0, \lambda_1 \gg \sigma_\phi$, where σ_ϕ is the characteristic phase fluctuation scale of the substrate;
- (iii) No continuous admissible path in Ω_{adm} connects B_0 to B_1 without crossing a region of vanishing admissibility measure.

Condition (ii) is the **classical stability condition**: the basins are deep enough that thermal or quantum phase fluctuations cannot spontaneously induce basin crossing. Condition (iii) is the **topological separation condition**: the basins are in different connected components of the admissibility landscape above the inter-basin saddle.

Classical determinism now emerges naturally. When conditions (i)–(iii) hold, any physical probe of the system will find it reliably in either B_0 or B_1 , never in a superposition (in the conventional sense), because the constraint geometry provides no admissible path connecting the basins except via an energy-costly restructuring. The apparent dichotomy of classical truth-values is a consequence of the *deep basin geometry*, not of any categorical feature of reality.

A **phase bit flip** is an admissible transformation $T_{\text{flip}} \in T_{\text{adm}}$ such that $T_{\text{flip}}(B_0) = B_1$ and $T_{\text{flip}}(B_1) = B_0$. Such a transformation must cross the inter-basin barrier

—the region of the admissibility landscape separating B_0 and B_1 —and its cost is:

$$C(T_{\text{flip}}) = \int_{\text{path}} \kappa(\Phi(s)) \, ds \quad (12.7)$$

where $\kappa(\Phi(s))$ is the admissibility curvature at the phase configuration $\Phi(s)$ along the inter-basin path, and s is a path parameter [PT3]. The energy cost of a bit flip is proportional to this path integral, anticipating the Landauer analysis of Section 10.

The **classical capacity** C_{class} of a phase system is defined as the maximum number of mutually distinguishable, classically stable basins:

$$C_{\text{class}} = \max\{ |M| : M \text{ is a mutually distinguishable set of phase basins satisfying conditions (i)–(iii)} \} \quad (12.8)$$

This quantity bounds the classical information-storing capacity of the phase system. It is always finite for systems with finite phase substrate volume (bounded Σ), and it is always a non-negative integer. The bound $C_{\text{class}} \leq \log_2 |\Omega_{\text{adm}}| / \lambda_{\min}$ shows that classical capacity is limited by the trade-off between total available configuration space and the minimum basin depth required for classical stability.

6. Quantum Bits Reinterpreted

The qubit—the quantum mechanical two-state system—is the cornerstone of quantum information theory. In the standard framework, a qubit is a unit vector in a two-dimensional complex Hilbert space $\mathcal{H}$, with superposition, interference, and entanglement as immediate consequences of the Hilbert space structure. Phase Information Theory must derive all of these

phenomena from the geometry of the admissible configuration space, without invoking Hilbert spaces as a primitive.

The key is the observation that the classical-bit conditions (i)–(iii) of Section 5 can fail in a graded manner. Specifically, condition (iii)—topological separation of basins—can fail continuously. When the admissibility boundary between the neighborhoods of B_0 and B_1 is not a sharp discontinuity but a region of partial admissibility, the two basins develop **overlapping basin neighborhoods**, giving rise to a continuous family of configurations intermediate between classical 0 and classical 1.

Define the **basin neighborhood** $N(B_\alpha) \subset \Omega_{\text{adm}}$ of a basin B_α as the set of configurations reachable from B_α by admissible transformations of cost at most one phase-unit:

$$N(B_\alpha) = \{ \Phi \in \Omega_{\text{adm}} : C(T_{B_\alpha \rightarrow \Phi}) \leq 1 \} \quad (12.9)$$

The **overlap depth** δ between two basin neighborhoods is defined as:

$$\delta(B_0, B_1) = \mu_{\text{adm}}(N(B_0) \cap N(B_1)) / \min(\mu_{\text{adm}}(N(B_0)), \mu_{\text{adm}}(N(B_1))) \quad (12.10)$$

where μ_{adm} is the natural measure on Ω_{adm} induced by the global constraint geometry [PT2].

Definition 12.4 (Phase Qubit)

A **phase qubit** is a pair of phase basins $\{B_0, B_1\} \subset \Omega_{\text{adm}}$ with overlap depth $\delta \in [0, 1]$. The pair is characterized by (B_0, B_1, δ) .

Proposition 6.1

The set of phase qubits forms a continuous family parameterized by overlap depth $\delta \in [0, 1]$, with classical bits at $\delta = 0$ and fully coherent qubits at $\delta = 1$. The classical limit recovers condition (iii) of Definition 12.3; the fully coherent limit corresponds to maximal basin neighborhood overlap.

Coherence is therefore not a quantum postulate in PIT; it is a structural property of the admissibility boundary geometry. When the boundary between $N(B_0)$ and $N(B_1)$ is soft (large overlap region), the system is highly coherent. When the boundary is sharp (zero overlap), the system is classical. The transition between these regimes—**decoherence**—corresponds to the geometric sharpening of the admissibility boundary, driven by global constraint reconfiguration as additional phase degrees of freedom couple to the system [PT8].

Quantum interference is reinterpreted as constructive or destructive overlap of basin neighborhood geometries. When two phase paths from B_0 to some target configuration Ψ pass through overlapping regions of $N(B_0)$ and $N(B_1)$, their admissibility amplitudes—defined as $\exp(-S_{\text{adm}}[\text{path}])$ where S_{adm} is the admissibility action [PT4]—can add constructively or cancel destructively, depending on the relative phase of the paths in the overlap region. This recovers the Born-rule interference pattern without invoking complex Hilbert space amplitudes as a primitive [PT10].

Measurement, as established in [PT6], is a constraint restructuring event that collapses the overlap region: an interaction with a macroscopic phase structure (the apparatus) dramatically sharpens the admissibility boundary, converting the phase qubit ($\delta > 0$) into a classical phase bit ($\delta = 0$) aligned with one or the other basin. The probabilities of each outcome are determined

by the admissibility measure of the respective basin neighborhoods at the moment of constraint restructuring, recovering the Born rule in the limit of large apparatus [PT6, PT10].

We now show that the density matrix formalism is an approximation valid in limiting regimes. Define the **phase density operator** ρ_Φ as the operator on the Hilbert space of basin-neighborhood functions $L^2(\Omega_{\text{adm}}, \mu_{\text{adm}})$ that projects onto the overlap region. In the limit $\delta \rightarrow 0$, $\rho_\Phi \rightarrow \text{diag}(p_0, p_1)$ —a classical mixture. In the limit $\delta \rightarrow 1$, $\rho_\Phi \rightarrow |\psi\rangle\langle\psi|$ where $|\psi\rangle$ is the equal-weight superposition state. The full density matrix formalism, including off-diagonal coherences, is recovered as:

$$\rho_{ij} = \int_{N(B_i) \cap N(B_j)} \exp(i S_{\text{adm}}[\Phi]) d\mu_{\text{adm}}(\Phi) \quad (12.11)$$

The off-diagonal elements ρ_{01} and ρ_{10} are precisely the integral of the admissibility phase factor over the overlap region, vanishing when $\delta = 0$ (no overlap) and reaching maximum magnitude when $\delta = 1$ (complete overlap). This shows that the density matrix is a derived construct in PIT, not a fundamental object.

7. Hybrid and Virtual Information Structures

The δ -parameterized family of Section 6 describes globally uniform phase qubits, in which the overlap depth is constant across the basin structure. Physical systems, however, commonly exhibit **heterogeneous basin profiles**: sub-regions of Ω_{adm} with different overlap depths, giving rise to hybrid and virtual information structures that do not fit cleanly into either the classical or the fully quantum regimes.

Define a **metastable phase attractor** as a phase basin $B_\alpha \subset \Omega_{\text{adm}}$ with finite basin depth λ_α that is comparable to the phase fluctuation scale σ_Φ : $1 \lesssim \lambda_\alpha/\sigma_\Phi \lesssim$

10 (in dimensionless units). Such a basin is stable over intermediate timescales but accessible to spontaneous admissible transitions on longer timescales. Metastable attractors are the natural substrates for **virtual information structures**—configurations that carry distinguishability for a finite time before transitioning to a more stable configuration.

A **virtual qubit** is a transient phase overlap configuration ($\delta_{\text{eff}}(t) > 0$) that persists for a characteristic coherence time τ_{coh} before the admissibility boundary sharpens ($\delta_{\text{eff}} \rightarrow 0$). The coherence time is determined by the rate at which global constraint reconfiguration sharpens the basin boundary, which in turn depends on the coupling of the local phase structure to the global substrate [PT8].

Hybrid classical-quantum information arises in systems with mixed δ profiles across sub-regions: some sub-regions of Ω_{adm} are in the classical regime ($\delta \approx 0$), while others are in the coherent regime ($\delta \approx 1$). The total information content is the sum of contributions from each sub-region, weighted by their admissibility measures.

Analog phase logic corresponds to continuously variable basin depth: instead of discrete $\{B_0, B_1\}$, the configuration space supports a continuum of basins $\{B_x\}_{x \in [0,1]}$, each with a continuously variable overlap depth $\delta(x, y)$ between B_x and B_y . Classical binary logic is recovered only in the limit of infinite basin separation ($\lambda_\alpha \rightarrow \infty$).

Theorem 7.1

Any finite-dimensional quantum information processing protocol admits a phase-structural implementation via controlled admissibility geometry. Specifically, any N -qubit unitary operation U on $\mathcal{H}^{\otimes N}$ corresponds to an admissible transformation $T_U \in T_{\text{adm}}$ that implements the same operation on

the phase basin structure of a corresponding N -fold phase qubit system.

Proof Sketch.

By the Solovay-Kitaev theorem and the universality of $\{H, T, \text{CNOT}\}$ gates, any N -qubit unitary can be approximated to arbitrary precision by a finite sequence of elementary gates from a universal set. Each elementary gate corresponds to a controlled modification of the admissibility boundary geometry: Hadamard gates correspond to equal-depth basin merging operations (setting $\delta = 1$ temporarily); phase gates correspond to admissibility phase rotations in the overlap region; CNOT gates correspond to conditional constraint couplings between two phase qubits. All such operations are elements of T_{adm} by construction (they preserve global admissibility). $\square$

The mechanism underlying **quantum error correction** in PIT is **controlled admissibility geometry**: by engineering phase basin structures with topological protection (basins separated by admissibility barriers that are topologically nontrivial [PT3, PT11]), errors—which correspond to inadvertent basin crossings—can be detected and corrected without measuring the information content directly, preserving the coherent phase structure throughout.

8. No-Cloning as a Structural Constraint

The quantum no-cloning theorem—the impossibility of perfectly duplicating an unknown quantum state—is one of the most celebrated results in quantum information theory [4, 5]. In the standard framework, it follows in two lines from the linearity of quantum mechanics: a hypothetical cloning operation would have to be linear and unitary, but no single linear map can clone non-

orthogonal states. In PIT, the theorem must be derived without invoking linearity or unitarity as axioms. We show that no-cloning follows directly from the non-separability of global phase constraints.

We first formalize the cloning problem in PIT. Suppose we have a single-copy phase system with admissible configuration space $\Omega_{\text{adm}}^{(1)}$, and a target blank system with space $\Omega_{\text{adm}}^{(2)}$. The combined system has admissible configuration space:

$$\begin{aligned} \Omega_{\text{adm}}^{(12)} &= \{ (\Phi_1, \Phi_2) \in \Omega_{\text{adm}}^{(1)} \times \Omega_{\text{adm}}^{(2)} : \\ &G_k^{(12)}[\Phi_1, \Phi_2] = 0 \quad \forall k \} \end{aligned} \quad (12.12)$$

where $G_k^{(12)}$ are the global constraints on the combined system. A **cloning map** would be a $T_{\text{clone}} \in T_{\text{adm}}^{(12)}$ such that:

$$T_{\text{clone}}(\Phi, \Phi_{\text{blank}}) = (\Phi, \Phi) \quad \forall \Phi \in \Omega_{\text{adm}}^{(1)} \quad (12.13)$$

for some fixed blank configuration $\Phi_{\text{blank}} \in \Omega_{\text{adm}}^{(2)}$.

Theorem 12.1 (Phase No-Cloning)

No admissible transformation $T_{\text{clone}} \in T_{\text{adm}}^{(12)}$ can perfectly duplicate an arbitrary admissible phase configuration. That is, for any $T \in T_{\text{adm}}^{(12)}$, there exists $\Phi \in \Omega_{\text{adm}}^{(1)}$ such that $T(\Phi, \Phi_{\text{blank}}) \neq (\Phi, \Phi)$.

Proof.

Suppose T_{clone} exists satisfying (12.13) for all $\Phi \in \Omega_{\text{adm}}^{(1)}$. Consider the global constraints $G_k^{(12)}$ on the combined system. In PIT, global constraints are generically non-separable: they do not factor as products of single-system constraints. Formally, the global constraint operator satisfies:

$$G_k^{(12)}[\Phi_1, \Phi_2] \neq G_k^{(1)}[\Phi_1] \otimes G_k^{(2)}[\Phi_2] \quad (12.14)$$

in general (this is the content of the Non-Separability Lemma of [PT2]). Now suppose Φ is a generic element of $\Omega_{\text{adm}}^{(1)}$ —one that is not in the T_{adm} -orbit of Φ_{blank} . The cloned configuration $(\Phi, \Phi) \in \Omega_{\text{adm}}^{(1)} \times \Omega_{\text{adm}}^{(2)}$ must satisfy all global constraints $G_k^{(12)}[\Phi, \Phi] = 0$. But since the combined system constraints are non-separable, the condition $G_k^{(12)}[\Phi, \Phi] = 0$ imposes cross-system correlations that are generically not satisfied by the “diagonal” configuration (Φ, Φ) when Φ is arbitrary. Specifically, for generic Φ distinct from Φ_{blank} , the cross-constraint:

$$G_k^{(\text{cross})}[\Phi, \Phi] = G_k^{(12)}[\Phi, \Phi] - (G_k^{(1)}[\Phi] \otimes G_k^{(2)}[\Phi]) \quad (12.15)$$

is generically nonzero, meaning $(\Phi, \Phi) \notin \Omega_{\text{adm}}^{(12)}$. Therefore $T_{\text{clone}}(\Phi, \Phi_{\text{blank}}) = (\Phi, \Phi)$ would map an element of $\Omega_{\text{adm}}^{(12)}$ to a point outside $\Omega_{\text{adm}}^{(12)}$, contradicting the requirement $T_{\text{clone}} \in T_{\text{adm}}^{(12)}$. Hence no such T_{clone} exists. $\square$

The standard quantum no-cloning theorem is recovered as a special case. In the $\delta = 1$ phase-qubit regime, the admissibility overlap structure is isomorphic to a two-dimensional complex Hilbert space (as shown in [PT10]). The global constraint non-separability in this regime corresponds precisely to the non-linearity of the requirement that a cloning map preserve inner products. The PIT proof thus strictly generalizes the Wootters-Zurek-Dieks proof [4, 5].

Approximate cloning is possible in PIT when the target configuration is restricted to a subspace of $\Omega_{\text{adm}}^{(1)}$ with reduced admissibility curvature. The maximal cloning fidelity F_{max} for arbitrary configurations is bounded by:

$$F_{\max} \leq 1 - \alpha / \kappa \quad (12.16)$$

where κ is the admissibility curvature at the basin boundary (defined precisely in [PT3]) and α is a universal PIT constant determined by the topology of $\Omega_{\text{adm}}^{(12)}$. In the limit $\kappa \rightarrow \infty$ (infinitely sharp admissibility boundary), $F_{\max} \rightarrow 1$, which would allow perfect cloning—but this limit corresponds to a degenerate phase substrate with vanishing coherence length, which is unphysical. The bound (12.16) is a testable prediction of PIT (see Section 15).

9. Information Capacity and Saturation

Having established what information is in PIT (phase distinguishability) and what constraints govern it (no-cloning, basin geometry), we now derive the fundamental limit on how much information a phase system can carry. This is the **phase information capacity**.

Definition 12.5 (Phase Information Capacity)

*The **phase information capacity** C_{phase} of a system with admissible configuration space Ω_{adm} is:*

C

phase

= log

2

(max

$M \subset \Omega$

adm

$|M|$)

(12.17)

where the maximum is taken over all mutually distinguishable subsets $M \subset \Omega_{\text{adm}}$.

Three geometric properties of Ω_{adm} bound C_{phase} from above:

(a) Phase coherence length λ_{coh} . This is the minimum spatial scale below which phase configurations cannot be distinguished by any admissible probe. It sets the minimum “grain size” of distinguishable configurations. Two configurations differing only on scales smaller than λ_{coh} are T_{adm} -equivalent and do not contribute independently to $|D|$. This gives a lower bound on the volume of each distinguishable cell in Ω_{adm} .

(b) Admissibility curvature κ . This is the curvature of the admissibility boundary at inter-basin saddle points [PT3]. High curvature κ implies sharply separated basins; low curvature implies shallow, overlapping basins. The inter-basin separation cost scales as $\kappa \lambda_{\text{coh}}^2$, bounding the number of basins that can fit in a given volume of Ω_{adm} .

(c) Topological protection density ρ_{top} . This is the density of topologically protected phase basins—those whose distinguishability is stable against all admissible perturbations of cost below a topological threshold [PT11]. Topologically protected basins contribute most stably to $|D|$; non-protected basins may merge under small perturbations.

Combining these three factors, the phase information capacity satisfies the upper bound:

$$C_{\text{phase}} \leq \log_2(\text{Vol}(\Omega_{\text{adm}}) / (\lambda_{\text{coh}}^d \times \kappa)) \quad (12.18)$$

where d is the effective dimension of the phase space and $\text{Vol}(\Omega_{\text{adm}})$ is the total admissibility volume. This bound (12.18) is the **Phase Information Capacity Bound**.

Phase information saturation is the condition $C_{\text{phase}} = \max$, i.e., when the bound (12.18) is saturated. At saturation, no further mutually distinguishable configuration can be added to the maximally distinguishable set D without violating at least one global constraint G_k . The system's phase structure is fully “packed” with distinguishable states.

We now show that the bound (12.18) subsumes the Bekenstein-Hawking holographic bound as a special case. For a spherically symmetric phase system of radius R , the admissibility volume is determined by the phase coherence length and the surface area $A = 4\pi R^2$ [PT11]. In the appropriate limit:

$$C_{\text{phase}} \leq A / (4 l_P^2) = S_{\text{BH}} / k_B \ln 2 \quad (12.19)$$

where l_P is the Planck length and S_{BH} is the Bekenstein-Hawking entropy [12, 13]. Thus the holographic bound is the special case of the Phase Information Capacity Bound under spherical symmetry, with the Planck length playing the role of the phase coherence length λ_{coh} . The holographic bound is not, in PIT, a mysterious fact about quantum gravity; it is a consequence of the geometry of the admissible configuration space in the regime of maximum phase curvature.

The Bekenstein bound [12]— $S \leq 2\pi RE/(\hbar c)$ —is similarly recovered as a sub-case of (12.18) in the non-relativistic limit, where energy E constrains the effective volume $\text{Vol}(\Omega_{\text{adm}})$ of accessible configurations via the phase-action bound [PT4].

10. Landauer’s Principle Reinterpreted

Landauer’s principle [14]—that the erasure of one bit of information dissipates at least $k_B T \ln 2$ of energy as heat—is one of the deepest connections between information theory and thermodynamics. In the standard derivation, the principle follows from the identification of Shannon entropy with thermodynamic entropy via Boltzmann’s constant, and from the Second Law. In PIT, where thermodynamic entropy is derived from phase basin statistics [PT8] and where information is phase distinguishability, the principle must be derived anew—and its derivation illuminates precisely where the energy cost originates.

In PIT, **information erasure** is defined as follows. Consider a system with two distinguishable basins B_0 and B_1 (a phase classical bit). Erasure is an admissible process $T_{\text{erase}} \in T_{\text{adm}}$ that maps both basins to a single final basin B_f :

$$T_{\text{erase}}(B_0) = T_{\text{erase}}(B_1) = B_f \quad (12.20)$$

This process: (1) merges two distinguishable phase basins into one; (2) reduces $|D|$ by at least 1; (3) necessarily disperses phase structure elsewhere in Ω_{adm} (since the global admissibility measure $\mu_{\text{adm}}(\Omega_{\text{adm}})$ is conserved under T_{adm} , the “deleted” distinguishability must be redistributed). The redistribution of phase structure into other degrees of freedom is what appears thermodynamically as heat dissipation.

The energy cost of erasure is determined by the cost of crossing the inter-basin admissibility barrier. From Section 5, the inter-basin barrier cost is $C(T_{\text{flip}}) = \int \kappa(\Phi(s)) ds$. For erasure, the relevant cost is the minimum-cost path that collapses B_0 and B_1 into B_f . Define κ_{basin} as the mean admissibility curvature along the optimal erasure path. Then:

$$\Delta E_{\text{erase}} \geq k_B T_{\text{eff}} \times \ln 2 \times \kappa_{\text{basin}} \quad (12.21)$$

where T_{eff} is the effective temperature of the phase substrate—defined in [PT8] as $T_{\text{eff}} = (\partial S_{\text{phase}} / \partial E)^{-1}$, with $S_{\text{phase}} = k_B \log|\Omega_{\text{adm}}|$ —and κ_{basin} is a dimensionless admissibility curvature factor.

Theorem 10.1 (Phase Landauer Principle)

Information erasure in Phase Theory is phase basin collapse: the merging of two distinguishable basins into one by an admissible process. The minimum energy dissipated in this process satisfies $\Delta E_{\text{erase}} \geq k_B T_{\text{eff}} \ln 2 \times \kappa_{\text{basin}}$, where κ_{basin} is the admissibility curvature at the basin boundary. In the classical thermodynamic limit ($\kappa_{\text{basin}} = 1$), this reproduces Landauer's bound exactly. The energy dissipation is a consequence of phase restructuring cost, not of information as a physical quantity.

The factor κ_{basin} is a PIT-specific correction to the Landauer bound. In conventional materials, $\kappa_{\text{basin}} \approx 1$, and the standard Landauer bound is recovered. In systems with engineered phase structure (e.g., topologically protected qubits [PT11]), κ_{basin} may deviate significantly from 1, providing a measurable correction (see Section 15.4 for the proposed experimental test).

Reversible computation is possible in PIT if and only if the admissible transformation T does not merge any two distinguishable basins. That is, T is reversible (in the information-theoretic sense) if and only if the induced map on the maximally distinguishable set D is a bijection. This recovers Bennett's result [15, 16] that reversible computation is possible without thermodynamic cost, while providing a precise geometric criterion: reversibility $\Leftrightarrow$ admissible-bijection on D . Importantly, the derivation does not

require thermodynamic axioms as input; thermodynamics is an output of the phase-structural analysis [PT8].

11. Information Flow Without Transport

Phase Information Theory must account for the apparent fact that information can be “transmitted” from one location to another. Since PIT denies that information is a substance, it cannot allow transport in the literal sense. We show that all phenomena that appear as information flow are, in fact, restructurings of the global distinguishability geometry—and that the rate of this restructuring is bounded by the phase propagation speed.

Define **apparent information flow** from location A to location B as follows. Let $D_A(t)$ be the set of phase configurations distinguishable by a probe localized at A at time t (more precisely: the set of configurations distinguishable using only admissible transformations whose support is concentrated at A). Then apparent information flow from A to B over the interval $[t_1, t_2]$ occurs when:

$$|D_B(t_2)| > |D_B(t_1)| \quad (12.22)$$

and this increase can be attributed to a phase constraint reconfiguration initiated at A. We show that all such increases are consequences of global phase restructuring—not of the transport of any phase-information substance.

Proposition 11.1: Let $T(t) \in T_{\text{adm}}$ be a one-parameter family of admissible transformations. The rate of change of $|D_B(t)|$ is bounded by:

$$d|D_B|/dt \leq v_{\text{phase}} \times \partial_{\text{boundary}} D_B \quad (12.23)$$

where v_{phase} is the phase propagation speed (the speed at which admissibility boundary changes propagate across Σ) and $\partial_{\text{boundary}} D_B$ is the surface area of the boundary of the B-local distinguishability region in Ω_{adm} .

Corollary 11.1 (Phase Signaling Limit)

No admissible process allows phase information restructuring to exceed the phase propagation speed v_{phase} . In the appropriate limit (flat phase substrate geometry, [PT11]), $v_{\text{phase}} = c$, the speed of light. This replaces the no-signaling theorems of standard quantum mechanics, which are derived consequences of no-cloning and linearity, with a structural phase-geometric bound.

The physical picture is as follows. When a sender at location A performs an admissible operation on the phase structure at A, the resulting change in the global constraint geometry propagates outward at v_{phase} . A receiver at location B cannot access the resulting change in D_B until the constraint restructuring reaches B. This is not information “traveling” from A to B; it is the propagation of a change in the *global geometry* of Ω_{adm} that happens to affect what is locally distinguishable at B. The distinction matters: in PIT, there is no substance traveling; there is only global constraint geometry evolving.

The Einstein-Podolsky-Rosen correlations and Bell non-locality are accommodated naturally in this picture. Entangled phase configurations have non-separable global constraints (Section 8); a constraint restructuring at A (a measurement) immediately reorganizes the global constraint structure—including the constraint structure at B—without any substance traveling from A to B. But this reorganization does not allow signaling because the change in D_B due to a constraint restructuring at A is indistinguishable from a change

due to local B processes, unless A and B compare their local distinguishability records via a classical phase propagation channel (at speed v_{phase}).

12. Entropy vs. Information

The relationship between entropy and information is one of the most conceptually fraught in all of physics. In Shannon theory, they are formally identical (Shannon entropy is “information” by definition). In thermodynamics, entropy is a macroscopic quantity related to microscopic probability distributions. In quantum mechanics, von Neumann entropy $S = -\text{Tr}(\rho \log \rho)$ measures the degree of mixedness of a quantum state, which is again identified with information. Phase Theory requires us to carefully distinguish these quantities and to derive their relationship from the phase-structural definitions.

From [PT8], the **phase entropy** S_{phase} of a system is defined as:

$$S_{\text{phase}} = k_B \log |\Omega_{\text{adm}}| \quad (12.24)$$

where $|\Omega_{\text{adm}}|$ is the total number of admissible configurations (or, in the continuous case, the logarithm of the admissibility volume). From Definition 12.2, the **phase information** is:

$$I_{\text{info}} = \log_2 |D| \quad (12.25)$$

where $D \subset \Omega_{\text{adm}}$ is the maximally distinguishable subset. These two quantities are related but distinct. Since $D \subset \Omega_{\text{adm}}$, we always have $|D| \leq |\Omega_{\text{adm}}|$, giving the fundamental inequality:

$$I_{\text{info}} \leq S_{\text{phase}} / (k_B \ln 2)$$

$$(12.26)$$

with equality if and only if every admissible configuration is mutually distinguishable from every other—the condition of **maximum information density**. Inequality (12.26) is the fundamental relation between phase information and phase entropy in PIT.

The ratio $R_{\text{info}} = |D| / |\Omega_{\text{adm}}|$ measures the **information efficiency** of the phase configuration space: what fraction of admissible configurations are mutually distinguishable. $R_{\text{info}} = 1$ corresponds to maximum information density (every admissible state can be distinguished from every other); $R_{\text{info}} \rightarrow 0$ corresponds to minimum information density (nearly all admissible states are T_{adm} -equivalent and thus indistinguishable).

This framework illuminates several physical phenomena:

High-entropy / low-information systems (large $|\Omega_{\text{adm}}|$, small $|D|$): Many admissible configurations exist, but most are T_{adm} -equivalent. Example: a thermalized gas has an enormous number of admissible microstates, but macroscopic probes can distinguish only a small number of macroscopic configurations. The collapse of $|D|$ relative to $|\Omega_{\text{adm}}|$ is precisely *thermalization*.

Low-entropy / high-information systems (small $|\Omega_{\text{adm}}|$, all or most states distinguishable): Few admissible configurations exist, and they are all mutually distinguishable. Example: a ground-state topologically ordered system has a small degenerate Hilbert space, all states of which are distinguishable by topological probes.

$$R_{\text{info}} = |D| / |\Omega_{\text{adm}}| = \exp(-\Delta S_{\text{phase}} / k_B)$$

$$(12.27)$$

where $\Delta S_{\text{phase}} = k_B \log(|\Omega_{\text{adm}}| / |D|)$ is the **entropy of indistinguishability**—the entropy attributable to T_{adm} -equivalence classes within Ω_{adm} .

Quantum error correction acts, in PIT, by increasing the ratio R_{info} . By engineering the basin structure so that errors (inadvertent basin crossings) map a state into a detectable but reversible orbit, error correction codes widen the effective basin structure and increase $|D|$ relative to $|\Omega_{\text{adm}}|$. The code subspace is precisely the high- R_{info} subspace within the full admissible space.

Fig. 12.1. *Schematic of admissible configuration space Ω_{adm} with the distinguishable subset D (shaded), for two limiting regimes. **Left:** Low-entropy / high-information regime: small Ω_{adm} , with nearly all admissible configurations mutually distinguishable ($D \approx \Omega_{\text{adm}}$, $R_{\text{info}} \approx 1$). Each basin is well separated from all others. **Right:** High-entropy / low-information regime: large Ω_{adm} , with most configurations T_{adm} -equivalent; only a small fraction $D \subset \Omega_{\text{adm}}$ is mutually distinguishable ($R_{\text{info}} \ll 1$). Thermalization corresponds to the growth of Ω_{adm} relative to D , collapsing the information efficiency.*

The von Neumann entropy $S_{\text{vN}} = -\text{Tr}(\rho \log \rho)$ is recovered from (12.27) in the δ -parameterized qubit regime. When the density matrix ρ is expressed as $\rho_{ij} = \int_{\text{overlap}} e^{iS_{\text{adm}}} d\mu_{\text{adm}}$ (equation 12.11), the eigenvalues of ρ are determined by the basin overlap geometry, and $-\text{Tr}(\rho \log \rho) = -\sum_i \lambda_i \log \lambda_i$ where λ_i are the eigenvalues of ρ_{ij} . This equals $\log_2 |D|$ in the maximally coherent limit, confirming consistency with Definition 12.2.

13. Black Holes and Information (Preview)

The black hole information paradox has been among the most stubbornly contested problems at the intersection of quantum mechanics and general

relativity for nearly half a century [17, 18, 19]. In its canonical form, the paradox arises because: (1) unitarity of quantum mechanics implies that information is never destroyed; (2) Hawking’s semiclassical calculation [17] implies that black holes evaporate into purely thermal (hence maximally mixed, hence minimally informative) radiation; (3) these two requirements are apparently inconsistent. Many proposed resolutions—firewalls, remnants, complementarity, ER=EPR [20]—have generated enormous controversy without achieving consensus.

Phase Information Theory dissolves this paradox by clarifying what “information loss” means in phase-structural terms. The resolution rests on the distinction between $|\Omega_{\text{adm}}|$ and $|D_{\text{external}}|$: the total admissible configuration space is never reduced by any admissible process (Proposition 13.0 below), but the **external distinguishability**—the number of configurations distinguishable by external probes—can decrease dramatically as a phase system approaches saturation.

In PIT, a **phase-saturated object** is a physical system whose phase information capacity (Definition 12.5) has been saturated: $C_{\text{phase}} = \max$, and no further distinguishable configurations can be added without violating global constraints. This is the PIT analog of a black hole. As a system approaches phase saturation:

$$|D_{\text{external}}| \rightarrow 0 \quad \text{as} \quad C_{\text{phase}} \rightarrow C_{\text{phase,max}} \quad (12.28)$$

while $|\Omega_{\text{adm}}|$ is preserved. The phase structure that was previously accessible to external probes is converted into internal phase structure—inaccessible to external observation but nonetheless real and admissible. The **external distinguishability horizon**, the boundary beyond which external probes lose distinguishability access, is the PIT analog of the event horizon.

Proposition 13.0 (Preview of Paper 13)

No admissible phase process reduces $|\Omega_{\text{adm}}|$. The apparent loss of information in a phase-saturated system corresponds to the collapse of the externally accessible distinguishable set D_{external} , not to the destruction of phase structure. Total phase information $I_{\text{info}} = \log|\Omega_{\text{adm}}|$ is conserved; external phase information $I_{\text{info,ext}} = \log|D_{\text{external}}|$ may decrease.

Hawking radiation [17], in this picture, is a phase decoherence process at the external distinguishability horizon: the admissibility boundary between the phase-saturated interior and the external substrate is not perfectly sharp (it has nonzero admissibility curvature κ), allowing slow leakage of phase constraint information from the interior to the exterior. This leakage appears to external observers as thermal radiation (because the leaked phase structure has maximal entropy from the external perspective) but carries internal phase correlations that encode the configuration of the interior.

Paper 13 of the Phase Theory series will formalize this account, deriving the Hawking temperature $T_{\text{H}} = \hbar\kappa_{\text{surface}}/(2\pi ck_{\text{B}})$ from the phase curvature κ_{surface} at the external distinguishability horizon, and proving that the total phase information $I_{\text{info}} = \log|\Omega_{\text{adm}}|$ is exactly conserved throughout the evaporation process. The paradox dissolves: information is not lost; it is encoded in internal phase structure that becomes accessible only as the external distinguishability horizon shrinks to zero—the PIT analog of complete evaporation.

14. Computational Implications

The phase-structural account of information has deep implications for the theory of computation. If computation is physical [21, 22], and if physical processes are phase-structural admissible transformations, then the theory of computation is a special case of the theory of admissible transformations on Ω_{adm} . We derive the principal consequences.

(a) Non-Boolean computation. Phase logic gates operate on continuous overlap depth $\delta \in [0, 1]$ rather than on discrete bit values $\{0, 1\}$. A phase gate G is an admissible transformation $T_G \in T_{\text{adm}}$ that maps $(\delta_{\text{in}}, \Phi_{\text{in}}) \rightarrow (\delta_{\text{out}}, \Phi_{\text{out}})$ with specified δ -transformation rules. Classical Boolean logic is the restriction of phase logic to $\delta = 0$; quantum logic is the restriction to $\delta = 1$; the full continuous family enables computation beyond binary. In particular, phase logic can implement graded-truth operations, analog-to-digital conversion at arbitrary precision, and probabilistic gates without any stochastic postulate.

(b) Coherence-based logic. Computation in the quantum regime ($\delta \approx 1$) proceeds via controlled phase overlap, as in quantum computation, but is derived from the phase geometry rather than postulated. Gate fidelity is determined by the admissibility curvature κ at the relevant basin boundaries; high κ gives high gate fidelity.

(c) Error correction via phase locking. Topologically protected phase basins—those whose distinguishability is stable against all admissible perturbations below a topological threshold—provide natural error correction without syndrome measurement. The topological protection density ρ_{top} determines the error correction capacity: a system with $\rho_{\text{top}} = 1$ (fully topologically protected) is immune to all sub-threshold errors. This is the phase-structural explanation for topological quantum codes (toric code, surface codes) [23].

(d) Computation without discrete state storage. Metastable phase attractors replace registers. A “register” in PIT is a metastable basin B_α with lifetime $\tau_\alpha \gg \tau_{\text{computation}}$. The information stored in B_α is the distinguishability of α from all other basins accessible to the computation; no discrete labeling is required.

We now define the Phase Computational Complexity classes.

Definition 12.6 (Phase-P and Phase-Q)

Phase-P is the class of decision problems solvable in polynomial time (in the input phase information I_{info}) by a sequence of admissible transformations from T_{adm} with $\delta = 0$ (phase-deterministic processes).

Phase-Q is the class of decision problems solvable in polynomial time by a sequence of admissible transformations from T_{adm} with $\delta \in [0, 1]$ (coherent phase processes).

Theorem 14.1

Phase-Q $\supseteq$ BQP. Any quantum algorithm implementable in BQP admits a phase-structural implementation in Phase-Q (by Theorem 7.1). Phase-Q may strictly contain BQP due to non-unitary effective dynamics arising from global constraint reconfiguration: admissible transformations in T_{adm} need not be unitary in the restricted Hilbert-space sense, enabling computation outside the unitary quantum circuit model.

Proof Sketch.

The inclusion $\text{Phase-Q} \supseteq \text{BQP}$ follows directly from Theorem 7.1: any BQP algorithm can be implemented by admissible transformations on a phase qubit system, so Phase-Q contains BQP. For the potential strict containment: Phase-Q allows admissible transformations that implement non-unitary maps on basin neighborhoods (specifically, maps that change δ while preserving global admissibility). Such transformations have no counterpart in the unitary quantum circuit model. Whether this extension gives computational power beyond BQP depends on whether global admissibility constraints can enforce correlations not achievable by local unitary operations—a question whose resolution requires more detailed study of the topology of Ω_{adm} in the relevant regime, to be addressed in a subsequent paper.

The implications for physical computation limits [21, 22] are also significant. Lloyd’s bound on the number of operations per second per unit energy [21] is recovered in PIT as a bound on the rate of admissible transformation of $|D|$, which is bounded by the phase propagation speed and the admissibility curvature (Proposition 11.1 and equation 12.21).

15. Experimental Consequences

A theoretical framework is scientifically valuable only insofar as it generates falsifiable empirical predictions that differ from those of existing theories. Phase Information Theory differs from standard quantum information theory in several testable respects. We enumerate the principal predictions and propose specific experimental protocols for each.

15.1 Temperature-Independent Information Limits

Standard quantum information theory predicts that information storage capacity is limited primarily by thermal fluctuations: as temperature $T \rightarrow 0$, capacity increases without bound (in the absence of other decoherence mechanisms). PIT predicts a different limit: capacity is bounded by $C_{\text{phase}} \leq$

$\log_2(\text{Vol}(\Omega_{\text{adm}}) / \lambda_{\text{coh}}^d k)$ regardless of temperature. The coherence length λ_{coh} is a property of the phase substrate geometry, not of thermal fluctuations; it remains nonzero even at $T = 0$.

Prediction 15.1: Information storage density in coherent quantum systems should saturate at a value determined by the phase coherence length, not by the thermal de Broglie wavelength, as temperature approaches zero. The saturation is observable as a plateau in the information density curve $I_{\text{info}}(T)$ below a critical temperature T_{coh} .

Proposed experiment: Measure the accessible number of distinguishable quantum states in a high-Q superconducting resonator or a topological qubit array as a function of temperature between 10 mK and 100 mK. Standard QIT predicts monotonic increase; PIT predicts saturation below $T_{\text{coh}} = \hbar v_F / (k_B \lambda_{\text{coh}})$, with the saturation value C_{phase} depending on the material phase coherence length measurable by independent transport experiments.

15.2 Distinguishability Collapse Without Decoherence

PIT predicts that systems near phase information saturation ($C_{\text{phase}} \rightarrow C_{\text{phase,max}}$) should exhibit a sudden collapse of the distinguishable channel count $|D|$ independent of environmental coupling. This is not decoherence in the standard sense (environmental entanglement causing off-diagonal density matrix elements to vanish); it is a structural consequence of the admissibility geometry reaching saturation.

Prediction 15.2: In a superconducting qubit array engineered to approach phase information saturation (by increasing qubit number at fixed coherence volume), the effective number of distinguishable computational basis states should show a non-monotonic dependence on qubit number: increasing up to a critical N_c , then collapsing abruptly, even in the absence of increased environmental coupling.

Proposed experiment: Fabricate superconducting transmon qubit arrays of increasing size $N = 10, 20, 30, 40, 50$, held at constant temperature and coupling strength. Measure the effective Hilbert space dimension (via quantum process tomography) as a function of N . Standard QIT predicts $D = 2^N$ monotonically; PIT predicts saturation and collapse at $N_c = \log_2(C_{\text{phase}})$ where C_{phase} is determined by the coherence length of the superconducting material.

15.3 Information Preservation Under Non-Unitary Dynamics

PIT predicts that global admissibility conservation preserves $|\Omega_{\text{adm}}|$ even under effective non-unitary evolution. In driven-dissipative quantum systems, the standard expectation is that information is lost to the environment. PIT predicts that the total phase information $I_{\text{info}} = \log|\Omega_{\text{adm}}|$ is conserved, and that apparent information loss is converted into internal phase structure inaccessible to standard probes.

Prediction 15.3: In a driven-dissipative quantum system (e.g., a coupled cavity-qubit system with engineered loss), quantum state reconstruction fidelity should be anomalously high compared to standard Lindblad master equation predictions, when reconstruction is performed using probes that access internal phase correlations (e.g., ancilla-assisted process tomography with environmental coupling).

Proposed experiment: Implement a driven-dissipative qubit in a microwave cavity. Compare state reconstruction fidelity using (a) standard homodyne detection (external probe only) and (b) ancilla-enhanced tomography that partially accesses the cavity mode (internal phase probe). PIT predicts that (b) recovers significantly more information than (a), with the difference quantified by the PIT formula $\Delta I = \log(|\Omega_{\text{adm}}|/|D_{\text{external}}|)$.

15.4 Landauer Corrections

The modified Landauer bound (equation 12.21) predicts a correction factor κ_{basin} to the standard $k_B T \ln 2$ bound. In conventional semiconductor devices, $\kappa_{\text{basin}} \approx 1$ and no correction is visible. In nano-scale devices with engineered phase structure (e.g., molecular magnets or atomic switches), κ_{basin} may deviate measurably from unity.

Prediction 15.4: The heat dissipated per erasure event in a nano-scale switch should deviate from $k_B T \ln 2$ by a factor κ_{basin} that is uniquely determined by the phase curvature of the material, measurable independently via neutron scattering or X-ray diffuse scattering.

Proposed experiment: Perform precision calorimetry of single-bit erasure events in (a) conventional CMOS switches at 300 K, (b) magnetic molecular junctions at 4 K, (c) topological insulator nanowire switches at 100 mK. Independently measure the phase coherence geometry via scattering experiments. PIT predicts $\Delta E_{\text{erase}} = k_B T \kappa_{\text{basin}} \ln 2$, with κ_{basin} materially dependent and predictable from the independent phase geometry measurement.

15.5 Phase No-Cloning Bound

The approximate cloning fidelity bound (equation 12.16) predicts $F_{\text{max}} = 1 - \alpha/\kappa$, where κ is the admissibility curvature and α is a universal PIT constant. In standard quantum information theory, the optimal cloning fidelity for an unknown qubit is $F = 5/6$ [24]. PIT predicts a correction to this value that depends on the physical implementation (specifically, on κ for the physical qubit).

Prediction 15.5: Optimal qubit cloning experiments performed on physical qubits with varying coherence geometry (e.g., different types of superconducting qubits: transmon, fluxonium, charge qubit) should yield cloning fidelities that deviate from the universal $5/6$ bound by

implementation-dependent corrections $F_{\max} = 5/6 - \alpha/\kappa_{\text{impl}}$, where κ_{impl} is the admissibility curvature of the specific qubit geometry.

Proposed experiment: Implement the Buzek-Hillery optimal quantum cloning machine [24] using (a) transmon qubits, (b) fluxonium qubits, and (c) photonic qubits, each with well-characterized coherence geometry. Measure achieved cloning fidelity for each implementation. Standard QIT predicts identical $F = 5/6$ for all; PIT predicts deviations proportional to $1/\kappa_{\text{impl}}$.

16. Comparison with Existing Frameworks

We now situate Phase Information Theory systematically within the landscape of existing information-theoretic frameworks, showing in each case how PIT extends, subsumes, or corrects the existing approach.

(a) Shannon Information Theory [1]. Shannon’s theory defines information as the reduction of uncertainty over a probability distribution on a finite alphabet. It is operationally precise but ontologically silent: it tells us how much information there is, not what information is. PIT contains Shannon’s theory as the limiting case in which: (i) the phase basins are classically stable ($\delta = 0$, condition (iii) of Definition 12.3); (ii) the admissibility measure μ_{adm} over basins defines a probability distribution $\{p_i\}$ by $p_i = \mu_{\text{adm}}(B_i) / \mu_{\text{adm}}(\Omega_{\text{adm}})$; (iii) Shannon entropy $H = -\sum p_i \log p_i$ is recovered from $I_{\text{info}} = \log |D|$ via equation (12.5). PIT adds the ontological grounding that Shannon theory lacks: probability arises from admissibility measure, not from an independent axiom.

(b) Quantum Information Theory (QIT) [2, 3]. QIT extends Shannon theory to quantum states, with von Neumann entropy, the Holevo bound, and quantum channels as its central objects. It takes the quantum state (density matrix) and Hilbert space as primitives. PIT contains QIT as the limiting case of the δ -parameterized family in the coherent regime ($\delta = 1$): density matrices

are derived from the phase overlap integral (12.11), the Holevo bound is recovered from the phase information capacity bound (12.18), and quantum channels are admissible transformations $T \in T_{\text{adm}}$ in the coherent phase qubit regime. PIT provides the derivational foundation that QIT takes as an axiom.

(c) It-from-Bit (Wheeler) [7]. Wheeler proposed that physical reality is constituted by binary observational outcomes. PIT inverts this priority: phase structure is fundamental, and bits are derived as the classical limit ($\delta = 0$) of phase distinguishability. The “bit” is not ontologically basic; it is a feature of the geometry of Ω_{adm} . Moreover, PIT does not require an observer to constitute physical facts; phase distinguishability is defined independently of any observation or question-answering.

(d) Algorithmic Information Theory (AIT) [6]. Kolmogorov complexity defines the information content of a string as the length of the shortest program that generates it. In PIT, the analogous quantity is the minimum admissibility-preserving transformation cost $C(T_{\Phi_1 \rightarrow \Phi_2})$ connecting two configurations. The complexity of a configuration Φ relative to a reference Φ_0 is: $K_{\text{phase}}(\Phi \mid \Phi_0) = \min_{T \in T_{\text{adm}} : T(\Phi_0) = \Phi} C(T)$. This recovers Kolmogorov complexity in the limit of a universal phase computation model, without requiring an uncomputable Turing machine as a primitive.

(e) Resource Theories in QIT [25]. Resource theories (of entanglement, coherence, asymmetry, magic, etc.) define what can be done for free and what requires a resource, providing an operational framework for quantum information processing. In PIT, all resource theories are unified: entanglement is non-separable admissibility constraints (Section 8); coherence is basin overlap depth δ (Section 6); asymmetry is the absence of a T_{adm} -symmetry; thermodynamic resources are admissibility measure asymmetries [PT8]. PIT is a meta-framework that recovers all existing resource theories as special cases of the distinguishability geometry.

(f) Holographic Information Theory (AdS/CFT) [26, 27, 28].

Holographic duality identifies the information content of a bulk region with the entanglement entropy of its boundary [26, 27]. PIT predicts holographic bounds as a consequence of phase saturation: when a d -dimensional phase system saturates its capacity C_{phase} , the maximally distinguishable set D is encoded on the $(d-1)$ -dimensional boundary of Σ (because the interior is fully phase-saturated and adds no further distinguishability). This gives the Ryu-Takayanagi formula [27] as a geometric identity on Ω_{adm} , without invoking string theory, extra dimensions, or the specific AdS/CFT correspondence as a starting point.

Framework	Information Defined As	Ontological Primitive	PIT Relationship	PIT Correction
Shannon IT	Reduction of uncertainty; $H = -\sum p \log p$	Probability distribution over alphabet	Special case ($\delta = 0$, uniform basins)	Ontological grounding via admissibility measure; no epistemic primitives
Quantum IT	Von Neumann entropy; $S = -\text{Tr}(\rho \log \rho)$	Quantum state (Hilbert space, density matrix)	Special case ($\delta = 1$ limit)	Hilbert space derived from phase geometry; state as emergent, not primitive
It-from-Bit	Binary observational outcomes constitute physical facts	Observation / binary question-answer	Ontological inversion: phase is primary, bit is derived	Observation is a phase process, not a primitive; bits are classical phase limits
Algorithmic IT	Shortest program length (Kolmogorov complexity)	Universal Turing machine	Generalized by minimum admissibility transformation cost	Computability issue avoided; complexity is physical, not Turing-machine-relative
Resource Theories	Operationally: what can be	Free operations	Unified: all resources are	Single framework

Framework	Information Defined As	Ontological Primitive	PIT Relationship	PIT Correction
	done freely vs. at a cost	(LOCC, thermal ops, etc.)	distinguishability geometry features	subsumes all resource theories via δ -parameterization
Holographic IT	Boundary entanglement entropy = bulk information	AdS/CFT correspondence, string theory	Phase saturation recovers holographic bounds	No string theory required; holography is a consequence of phase capacity geometry

17. Conclusion

This paper has developed Phase Information Theory (PIT) as a complete foundational reformulation of information theory grounded in the phase-structural framework of the Phase Theory series. The central claim, established through formal derivation rather than mere analogy, is that information is not an independent ontological primitive but an emergent structural property of the admissible configuration space Ω_{adm} : specifically, the log-cardinality of the maximally mutually distinguishable subset $D \subset \Omega_{\text{adm}}$.

We summarize the four key conceptual advances of this paper.

First: Information is emergent, not primitive. Theorem 2.1 (Phase Symbol Neutrality) establishes that no information quantity can be defined from local phase values alone. Definition 12.2 derives information from the global geometry of Ω_{adm} via distinguishability relations. This removes information from the list of physical primitives alongside the eliminations of intrinsic randomness [PT5], wavefunction collapse [PT6], and fundamental particles and fields [PT7–PT9]. The project of Papers 1–12 has been to derive

all of physics from phase structure alone; the treatment of information here completes the conceptual core of that project.

Second: Classical and quantum information are unified under a single δ -parameterized framework. The overlap depth $\delta \in [0, 1]$ of Proposition 6.1 provides a continuous interpolation between classical bits ($\delta = 0$) and fully coherent quantum bits ($\delta = 1$), with hybrid and virtual information structures occupying the interior. This unification is not merely formal; it provides a geometric explanation for why quantum information is “more powerful” than classical information (larger basin-neighborhood overlap supports richer computational structures) and why coherence is fragile (small perturbations of Ω_{adm} can sharpen basin boundaries and reduce δ).

Third: No new ontology is required. Classical bits, quantum bits, density matrices, Shannon entropy, Holevo bounds, no-cloning, Landauer’s principle, holographic bounds, and quantum error correction are all derived from the single structure: the global admissibility geometry of Ω_{adm} , already established in Papers 1–3 of the series. PIT does not add new axioms; it draws out implications of the framework already in place.

Fourth: Information paradoxes dissolve into phase-structural statements. The black hole information paradox becomes the statement that $|D_{\text{external}}|$ can decrease while $|\Omega_{\text{adm}}|$ is preserved (Section 13). The Landauer paradox (why does erasing information cost energy?) becomes the statement that basin collapse requires crossing an admissibility barrier (Section 10). The no-cloning puzzle (why can quantum states not be copied?) becomes the statement that global phase constraints are non-separable (Theorem 12.1). In each case, the apparent mystery arose from misidentifying information as a substance; the phase-structural reformulation removes the mystery by removing the misidentification.

The philosophical coda that closes Paper 12 is not decorative but structural: *information is not stored in reality. Reality is structured so that distinctions can exist.* The admissible configuration space Ω_{adm} , with its basin geometry, overlap structure, and topological protection, is what makes it possible for some configurations to be different from others in a way that is stable, accessible, and physically meaningful. Information is this possibility, measured logarithmically.

Paper 13 of the Phase Theory series will formalize the treatment of black holes previewed in Section 13: deriving Hawking radiation as a phase decoherence process at the external distinguishability horizon, proving exact conservation of $|\Omega_{\text{adm}}|$ throughout black hole evaporation, and resolving the information paradox completely within the phase-structural framework. The results of the present paper—particularly Definition 12.2, Theorem 12.1, and the Phase Information Capacity Bound (12.18)—will serve as the principal technical tools in that derivation.

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